

Results Supplement

Predicting NMR Chemical Shifts from Sequence

This supplement records the full-detail calculated-shielding, chemical-shift development, trajectory and static-external tables supporting the Results landed at buffer commit `edcaf7f366ae2579541f4e1d6b3dce5fef7eb218`. All shift-model errors are calculated from exact assignments and reported in parts per million (ppm). H, HA, CA, CB, C and N are reported separately; HA contains HA, HA2 and HA3. No pooled H/C/N score or selector loss is reported.

For calculated shielding, T_0 is the isotropic component. The five T_2 values use an orthonormal symmetric-traceless basis; their RMS and mean norms are therefore the corresponding tensor Frobenius-error norms.

The static and first-retained-frame **ShiftTfnModel** fits are independent development routes. **ShiftMDModel** is fitted only to trajectory inputs using the fixed selected first-retained-frame development parent. The external **RefDbTestSmallSet** and **RefDbTestLargeSet** comparisons instead use four independently refitted static **Shift-TfnModel** source-contrast conditions: Plain, ORCA transfer, MOPAC and AIMNet2. Those condition labels are not aliases for the canonical development forms **MinimalCore**, **Aimnet2Embedding**, **MinimalMopac** and **All(Aimnet2+Mopac)**. No first-retained-frame or trajectory model is applied to a static external structure.

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1 Calculated-Shielding Results

ShieldingTfnModel condition	Atom	AFShieldingSet: fitting (521 proteins)			AFShieldingSet: selector (25 proteins)			AFShieldingSet: protected (79 proteins)		
		T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine
MinimalCore	C α	0.484	91.9%	0.9968	0.648	90.8%	0.9960	0.626	90.9%	0.9959
	C β	0.444	90.6%	0.9955	0.654	88.8%	0.9938	0.655	89.2%	0.9940
	C'	0.367	96.1%	0.9994	0.518	96.1%	0.9994	0.550	95.9%	0.9993
	O	2.364	96.2%	0.9995	2.932	95.9%	0.9994	3.186	95.9%	0.9994
	N	0.937	96.3%	0.9991	1.384	95.7%	0.9988	1.304	95.8%	0.9989
	H α	0.076	90.1%	0.9942	0.093	88.0%	0.9916	0.087	88.7%	0.9925
	H ^N	0.135	92.9%	0.9975	0.152	92.0%	0.9967	0.153	92.0%	0.9970
Aimnet2Embedding	C α	0.426	92.5%	0.9973	0.614	91.3%	0.9966	0.613	91.2%	0.9962
	C β	0.383	91.4%	0.9962	0.626	89.2%	0.9941	0.677	89.7%	0.9945
	C'	0.330	96.4%	0.9995	0.539	96.1%	0.9994	0.563	96.0%	0.9994
	O	2.158	96.3%	0.9995	3.156	95.9%	0.9994	3.097	95.9%	0.9994
	N	0.842	96.4%	0.9992	1.320	95.6%	0.9985	1.252	95.8%	0.9989
	H α	0.071	90.7%	0.9950	0.088	88.7%	0.9928	0.090	89.1%	0.9931
	H ^N	0.120	93.5%	0.9979	0.149	92.5%	0.9972	0.146	92.5%	0.9973
MinimalMopac	C α	0.406	92.6%	0.9973	0.574	91.6%	0.9967	0.594	91.2%	0.9960
	C β	0.404	91.5%	0.9963	0.621	89.5%	0.9946	0.667	89.4%	0.9944
	C'	0.323	96.4%	0.9995	0.490	96.3%	0.9995	0.516	95.8%	0.9994
	O	2.234	96.5%	0.9996	3.036	96.4%	0.9995	3.266	96.2%	0.9995
	N	0.884	96.5%	0.9993	1.193	96.0%	0.9987	1.346	95.7%	0.9989
	H α	0.065	91.6%	0.9958	0.083	89.9%	0.9939	0.083	89.4%	0.9937
	H ^N	0.114	93.8%	0.9981	0.138	93.0%	0.9975	0.142	92.8%	0.9974
All(Aimnet2+Mopac)	C α	0.378	93.3%	0.9978	0.563	92.2%	0.9972	0.619	91.4%	0.9964
	C β	0.321	92.2%	0.9969	0.564	89.9%	0.9950	0.625	89.3%	0.9944
	C'	0.301	96.7%	0.9996	0.498	96.3%	0.9995	0.511	96.1%	0.9994
	O	1.879	96.8%	0.9996	2.771	96.5%	0.9995	2.990	96.3%	0.9995
	N	0.804	96.7%	0.9994	1.269	96.1%	0.9989	1.293	96.0%	0.9991
	H α	0.060	91.8%	0.9961	0.078	89.6%	0.9938	0.081	89.6%	0.9939
	H ^N	0.105	94.2%	0.9983	0.144	93.1%	0.9976	0.132	93.1%	0.9977

T_0 RMSE is in shielding ppm. T_2 recovery is $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{zero tensor}}$; cosine is the mean normalized dot product.

Table S1: Backbone shielding performance across the three **AFShieldingSet** partitions.

ShieldingTfnModel condition	Atom	TrajectoryTest1P9JSet (751 frames)			TrajectoryTest1L2YSet (1,501 frames)			TrajectoryTest5AWLSet (5,001 frames)		
		T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine	T_0 RMSE	T_2 rec.	cosine
MinimalCore	C α	3.819	69.2%	0.9517	3.813	69.8%	0.9484	3.983	67.4%	0.9445
	C β	3.525	63.7%	0.9252	3.007	68.6%	0.9499	3.276	64.5%	0.9355
	C'	4.289	89.1%	0.9953	4.108	89.8%	0.9958	4.214	89.4%	0.9954
	N	6.637	85.9%	0.9879	6.366	86.4%	0.9792	6.249	85.5%	0.9432
	H α	0.310	78.0%	0.9717	0.280	79.9%	0.9770	0.289	78.7%	0.9779
	H ^N	0.425	85.3%	0.9894	0.387	86.7%	0.9924	0.428	84.3%	0.9873
Aimnet2Embedding	C α	3.787	71.1%	0.9578	3.698	70.9%	0.9545	3.839	70.6%	0.9575
	C β	3.739	65.0%	0.9304	2.882	68.5%	0.9510	3.154	66.2%	0.9422
	C'	4.304	89.4%	0.9960	4.249	89.7%	0.9962	4.365	89.8%	0.9961
	N	6.552	86.0%	0.9889	6.655	86.9%	0.9855	9.600	85.7%	0.9553
	H α	0.279	78.6%	0.9726	0.265	79.5%	0.9754	0.270	79.3%	0.9782
	H ^N	0.427	85.2%	0.9890	0.397	86.0%	0.9915	0.403	84.8%	0.9881
MinimalMopac	C α	3.901	55.3%	0.9029	3.525	71.8%	0.9563	3.765	70.2%	0.9557
	C β	4.244	51.9%	0.8931	2.947	69.7%	0.9546	3.170	66.6%	0.9404
	C'	4.964	78.5%	0.9845	4.518	89.0%	0.9956	4.529	88.1%	0.9946
	N	8.827	77.1%	0.9757	6.970	86.3%	0.9830	6.659	85.4%	0.9487
	H α	0.427	55.1%	0.8904	0.276	80.0%	0.9784	0.322	78.3%	0.9749
	H ^N	0.785	60.9%	0.9558	0.458	85.4%	0.9916	0.493	84.3%	0.9869
All(Aimnet2+Mopac)	C α	3.877	58.2%	0.9154	3.813	71.7%	0.9574	3.885	70.6%	0.9569
	C β	4.089	51.1%	0.8883	2.983	70.3%	0.9575	3.321	68.6%	0.9470
	C'	4.783	81.2%	0.9863	4.399	90.1%	0.9964	4.393	89.9%	0.9962
	N	7.562	79.7%	0.9819	6.115	86.6%	0.9812	6.718	85.7%	0.9569
	H α	0.478	53.2%	0.8774	0.269	80.5%	0.9788	0.282	79.4%	0.9777
	H ^N	0.703	67.9%	0.9424	0.414	87.2%	0.9928	0.464	86.3%	0.9907

T_0 RMSE is in shielding ppm. T_2 recovery is $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{zero tensor}}$; cosine is the mean normalized dot product.

Table S2: Backbone shielding performance on the three trajectory test sets.

1.1 Complete selector screen

No.	ShieldingTfnModel screen condition	T_0 RMSE (ppm)						
		C α	C β	C'	O	N	H α	H N
1	Structure only	1.087	0.995	0.925	5.422	2.125	0.137	0.306
2	AIMNet2 embedding	0.886	0.918	0.775	4.982	1.882	0.117	0.231
3	Embedding + AIMNet2 electrostatics	0.864	0.880	0.740	4.888	1.969	0.117	0.208
4	Embedding + AIMNet2 response	0.903	0.883	0.783	4.805	1.880	0.116	0.230
5	Embedding + AIMNet2 energy/dispersion	0.888	0.934	0.784	4.972	1.919	0.117	0.229
6	All AIMNet2	0.889	0.908	0.774	4.569	1.937	0.118	0.209
7	MOPAC core, no AIMNet2	0.972	0.973	0.792	5.035	1.888	0.118	0.268
8	Embedding + MOPAC atomic state	0.881	0.907	0.773	5.110	1.908	0.118	0.229
9	Embedding + MOPAC density/bonding	0.877	0.866	0.744	4.531	1.906	0.120	0.206
10	Embedding + MOPAC electrostatics	0.915	0.849	0.766	4.658	1.860	0.126	0.215
11	Embedding + MOPAC peptide C/O	0.893	0.902	0.743	4.765	2.038	0.120	0.217
12	Embedding + MOPAC core	0.892	0.873	0.737	4.460	1.731	0.111	0.211
13	Embedding + ring current	0.939	0.978	0.774	5.002	1.895	0.122	0.217
14	Embedding + H-bond geometry	0.910	0.945	0.798	4.868	1.923	0.120	0.223
15	Embedding + SASA	0.914	0.901	0.779	5.204	1.944	0.118	0.228
16	Embedding + fixed McConnell	0.873	0.889	0.728	4.249	1.946	0.114	0.226
17	Embedding + APBS	0.903	0.955	0.805	5.146	1.841	0.117	0.227
18	Embedding + combined environment	0.861	0.906	0.763	4.363	2.078	0.119	0.222
19	Full, embedding-only AIMNet2	0.830	0.849	0.730	4.440	1.764	0.108	0.200
20	Full, all AIMNet2	0.874	0.832	0.720	4.554	1.741	0.111	0.199

Table S3: Complete 20-condition **AFShieldingSet** selector screen (25 proteins): backbone T_0 RMSE (ppm). These screen labels describe input ablations and are not aliases for the four canonical project forms.

No.	ShieldingTfnModel screen condition	T_2 recovery						
		$C\alpha$	$C\beta$	C'	O	N	$H\alpha$	H^N
1	Structure only	85.2%	82.2%	93.5%	93.6%	93.5%	85.0%	89.3%
2	AIMNet2 embedding	86.6%	84.1%	94.2%	93.9%	94.0%	85.8%	90.2%
3	Embedding + AIMNet2 electrostatics	86.6%	83.8%	94.3%	94.2%	94.0%	86.4%	90.0%
4	Embedding + AIMNet2 response	86.6%	84.3%	94.4%	94.1%	94.0%	86.0%	90.3%
5	Embedding + AIMNet2 energy/dispersion	86.7%	84.5%	94.2%	94.2%	94.1%	86.4%	90.4%
6	All AIMNet2	86.7%	84.0%	94.3%	94.3%	94.1%	86.9%	90.2%
7	MOPAC core, no AIMNet2	86.8%	84.5%	94.6%	94.5%	94.1%	86.1%	90.1%
8	Embedding + MOPAC atomic state	86.6%	84.1%	94.2%	94.1%	94.0%	85.9%	90.3%
9	Embedding + MOPAC density/bonding	87.2%	84.8%	94.6%	94.6%	94.4%	86.7%	90.2%
10	Embedding + MOPAC electrostatics	86.8%	84.3%	94.4%	94.1%	93.7%	86.5%	90.1%
11	Embedding + MOPAC peptide C/O	86.6%	83.5%	94.2%	94.3%	94.1%	85.5%	90.0%
12	Embedding + MOPAC core	87.5%	84.7%	94.7%	94.8%	94.2%	86.5%	90.6%
13	Embedding + ring current	86.1%	83.2%	94.2%	94.2%	93.8%	85.5%	89.9%
14	Embedding + H-bond geometry	86.6%	83.8%	94.1%	94.1%	94.0%	86.1%	90.0%
15	Embedding + SASA	86.7%	84.4%	94.2%	94.1%	93.9%	86.1%	90.3%
16	Embedding + fixed McConnell	86.3%	83.7%	94.4%	94.1%	93.9%	85.9%	90.0%
17	Embedding + APBS	86.6%	84.3%	94.2%	94.0%	94.0%	85.9%	89.9%
18	Embedding + combined environment	85.7%	82.9%	93.6%	94.0%	93.9%	85.6%	89.5%
19	Full, embedding-only AIMNet2	86.7%	84.4%	94.5%	94.7%	94.0%	86.6%	90.2%
20	Full, all AIMNet2	86.7%	83.9%	94.8%	94.5%	94.1%	87.6%	90.7%

Table S4: Complete 20-condition **AFShieldingSet** selector screen (25 proteins): backbone T_2 recovery relative to the zero-tensor baseline. These screen labels describe input ablations and are not aliases for the four canonical project forms.

2 Chemical-Shift Development Controls

Input form	ShiftTfnModel route	H	HA	CA	CB	C	N
MinimalCore	static ShiftTfnModel	0.368	0.188	0.755	0.898	1.134	1.983
MinimalCore	first-retained-frame ShiftTfn-Model	0.446	0.228	0.968	1.048	1.251	2.244
MinimalMopac	static ShiftTfnModel	0.358	0.184	0.746	0.873	1.132	1.943
MinimalMopac	first-retained-frame ShiftTfn-Model	0.440	0.230	0.941	1.033	1.275	2.475
Aimnet2Embedding	static ShiftTfnModel	0.399	0.211	0.879	0.977	1.235	2.394
Aimnet2Embedding	first-retained-frame ShiftTfn-Model	0.440	0.230	0.978	1.048	1.215	2.230
All(Aimnet2+Mopac)	static ShiftTfnModel	0.401	0.205	0.876	1.002	1.215	2.521
All(Aimnet2+Mopac)	first-retained-frame ShiftTfn-Model	0.451	0.225	0.995	1.067	1.198	2.352

Table S5: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of **RefDbSmallSet**. Static fits used **RefDbSmallTleapSet**; first-retained-frame fits used **RefDbSmallGromacsSet**.

Model condition	H	HA	CA	CB	C	N
static MinimalCore ShiftTfnModel	0.368	0.188	0.755	0.898	1.134	1.983
static MinimalCore ShiftTfnModel without OpenFold3	0.381	0.194	0.870	0.920	1.165	2.012
static MinimalCore ShiftTfnModel without shielding-derived information	0.434	0.237	0.840	0.957	1.226	2.640
first-retained-frame MinimalCore ShiftTfnModel	0.446	0.228	0.968	1.048	1.251	2.244
first-retained-frame MinimalCore ShiftTfn-Model without shielding-derived information	0.490	0.285	1.009	1.076	1.313	2.837

Table S6: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of **RefDbSmallSet**; static and first-retained-frame rows used **RefDbSmallTleapSet** and **RefDbSmallGromacsSet**, respectively. The shielding-information control jointly replaced the shielding-trained chemical encoder with a randomly initialised encoder and removed predicted T_0 , T_2 magnitude and T_2 shape, ring current and the five McConnell inputs.

ShiftMDModel condition	H	HA	CA	CB	C	N
Conditioned full	0.4132	0.2091	0.8651	0.9861	1.2004	2.1003
without frozen-parent response	0.4061	0.2078	0.8600	0.9746	1.1775	2.1003
without scalar structure and electronic quantities	0.4122	0.2079	0.8651	0.9861	1.2004	2.0995
without vector and rank-2 invariants	0.4121	0.2087	0.8651	0.9861	1.2004	2.0979
without shielding and physical values	0.4132	0.2095	0.8651	0.9861	1.2004	2.1001
without torsion and rotamer quantities	0.4140	0.2092	0.8651	0.9861	1.2004	2.1006
without aligned atomic motion	0.4132	0.2088	0.8651	0.9861	1.2004	2.1001
without support and DSSP state	0.4135	0.2090	0.8651	0.9861	1.2004	2.1004

Table S7: Exact-assignment backbone RMSE (ppm) on the 22 selector proteins of **RefDbSmallGromacsSet**. Each row refitted the seed-0 conditioned-full **ShiftMDModel** after suppressing the named input family; its selected first-retained-frame **ShiftTfnModel** parent remained fixed.

3 Shielding-to-Shift Transfer Factorial

Arm	Encoder start	Nine-value route	Role-balanced loss		H	HA	CA	CB	C	N
static ShiftTfnModel	Fresh	Zero	0.411 $\pm$ 0.007	0.447 $\pm$ 0.002	0.254 $\pm$ 0.009	0.903 $\pm$ 0.025	1.006 $\pm$ 0.030	1.040 $\pm$ 0.010	2.673 $\pm$ 0.038	
static ShiftTfnModel	Fresh	Supplied	0.388 $\pm$ 0.002	0.420 $\pm$ 0.004	0.213 $\pm$ 0.003	0.903 $\pm$ 0.009	1.000 $\pm$ 0.013	1.025 $\pm$ 0.010	2.707 $\pm$ 0.095	
static ShiftTfnModel	Shielding-pretrained	Zero	0.340 $\pm$ 0.004	0.378 $\pm$ 0.004	0.192 $\pm$ 0.002	0.817 $\pm$ 0.017	0.889 $\pm$ 0.010	0.863 $\pm$ 0.015	2.311 $\pm$ 0.032	
static ShiftTfnModel	Shielding-pretrained	Supplied	0.339 $\pm$ 0.003	0.380 $\pm$ 0.004	0.195 $\pm$ 0.003	0.807 $\pm$ 0.005	0.899 $\pm$ 0.010	0.883 $\pm$ 0.019	2.358 $\pm$ 0.093	
first-retained-frame ShiftTfn-Model	Fresh	Zero	0.457 $\pm$ 0.001	0.477 $\pm$ 0.009	0.284 $\pm$ 0.006	1.068 $\pm$ 0.015	1.140 $\pm$ 0.012	1.144 $\pm$ 0.044	3.136 $\pm$ 0.053	
first-retained-frame ShiftTfn-Model	Fresh	Supplied	0.437 $\pm$ 0.003	0.460 $\pm$ 0.007	0.253 $\pm$ 0.004	1.073 $\pm$ 0.021	1.137 $\pm$ 0.011	1.126 $\pm$ 0.022	3.096 $\pm$ 0.087	
first-retained-frame ShiftTfn-Model	Shielding-pretrained	Zero	0.400 $\pm$ 0.004	0.437 $\pm$ 0.005	0.230 $\pm$ 0.003	0.985 $\pm$ 0.021	1.064 $\pm$ 0.008	1.044 $\pm$ 0.055	2.757 $\pm$ 0.022	
first-retained-frame ShiftTfn-Model	Shielding-pretrained	Supplied	0.396 $\pm$ 0.003	0.435 $\pm$ 0.003	0.230 $\pm$ 0.001	0.978 $\pm$ 0.009	1.063 $\pm$ 0.013	1.020 $\pm$ 0.005	2.709 $\pm$ 0.062	

Table S8: Three-seed shielding-to-shift transfer factorial on the 45 protected proteins of **RefDbSmallSet**: backbone RMSE (ppm) as mean $\pm$ sample SD. Static and first-retained-frame arms used **RefDbSmallTleapSet** and **RefDbSmallGromacsSet**, respectively. The role-balanced loss, shown only with RMSE, is dimensionless and is not selector loss.

Arm	Encoder start	Nine-value route	H	HA	CA	CB	C	N
static ShiftTfnModel	Fresh	Zero	0.325 ± 0.001	0.182 ± 0.006	0.680 ± 0.020	0.718 ± 0.023	0.795 ± 0.008	1.981 ± 0.037
static ShiftTfnModel	Fresh	Supplied	0.306 ± 0.005	0.157 ± 0.003	0.677 ± 0.013	0.713 ± 0.011	0.782 ± 0.008	2.007 ± 0.072
static ShiftTfnModel	Shielding-pretrained	Zero	0.274 ± 0.003	0.140 ± 0.001	0.608 ± 0.020	0.623 ± 0.006	0.652 ± 0.011	1.662 ± 0.023
static ShiftTfnModel	Shielding-pretrained	Supplied	0.275 ± 0.004	0.141 ± 0.001	0.601 ± 0.013	0.630 ± 0.004	0.662 ± 0.014	1.701 ± 0.076
first-retained-frame ShiftTfn-Model	Fresh	Zero	0.353 ± 0.008	0.204 ± 0.003	0.823 ± 0.014	0.822 ± 0.007	0.883 ± 0.039	2.344 ± 0.056
first-retained-frame ShiftTfn-Model	Fresh	Supplied	0.340 ± 0.005	0.185 ± 0.003	0.828 ± 0.015	0.818 ± 0.004	0.871 ± 0.018	2.313 ± 0.066
first-retained-frame ShiftTfn-Model	Shielding-pretrained	Zero	0.326 ± 0.006	0.169 ± 0.002	0.754 ± 0.024	0.768 ± 0.011	0.802 ± 0.048	2.047 ± 0.027
first-retained-frame ShiftTfn-Model	Shielding-pretrained	Supplied	0.325 ± 0.006	0.168 ± 0.000	0.745 ± 0.012	0.768 ± 0.015	0.781 ± 0.003	1.999 ± 0.060

Table S9: Three-seed shielding-to-shift transfer factorial on the 45 protected proteins of **RefDbSmallSet**: backbone MAE (ppm) as mean $\pm$ sample SD. Static and first-retained-frame arms used **RefDbSmallTleapSet** and **RefDbSmallGromacsSet**, respectively. The role-balanced loss, shown only with RMSE, is dimensionless and is not selector loss.

4 Trajectory Development and Protected Results

Model	Condition	Selected	H	HA	CA	CB	C	N
ShiftTfnModel	Frame 15 baseline	No	0.446	0.228	0.968	1.048	1.251	2.244
ShiftTfnModel	Uniform mean baseline	No	0.409	0.208	0.865	0.986	1.200	2.100
ShiftMDModel	Historical contrast full	No	0.431	0.214	0.895	0.990	1.212	2.194
ShiftMDModel	Conditioned first moment	No	0.408	0.208	0.865	0.986	1.200	2.099
ShiftMDModel	Conditioned first, second and ordered	No	0.408	0.208	0.865	0.986	1.200	2.100
ShiftMDModel	Conditioned full	Yes	0.413	0.209	0.865	0.986	1.200	2.100

Table S10: **RefDbSmallGromacsSet** selector (22 proteins): exact-assignment backbone RMSE (ppm) for the fixed **ShiftTfnModel** baselines and seed-0 **ShiftMDModel** trajectory designs. Selector loss is not printed.

Model	Condition	Selected	H	HA	CA	CB	C	N
ShiftTfnModel	Frame 15 baseline	No	0.332	0.160	0.726	0.703	0.865	1.731
ShiftTfnModel	Uniform mean baseline	No	0.301	0.148	0.640	0.631	0.817	1.610
ShiftMDModel	Historical contrast full	No	0.320	0.149	0.671	0.655	0.814	1.688
ShiftMDModel	Conditioned first moment	No	0.300	0.148	0.640	0.631	0.817	1.609
ShiftMDModel	Conditioned first, second and ordered	No	0.300	0.147	0.640	0.631	0.817	1.610
ShiftMDModel	Conditioned full	Yes	0.306	0.146	0.640	0.631	0.817	1.610

Table S11: **RefDbSmallGromacsSet** selector (22 proteins): exact-assignment backbone MAE (ppm) for the fixed **ShiftTfnModel** baselines and seed-0 **ShiftMDModel** trajectory designs. Selector loss is not printed.

Model	Route	Statistic	H	HA	CA	CB	C	N
ShiftTfnModel	Selected static	Fixed	0.369	0.196	0.811	0.884	0.889	2.375
ShiftTfnModel	Frame 15 baseline	Fixed	0.442	0.230	0.991	1.058	1.030	2.701
ShiftTfnModel	Uniform mean baseline	Fixed	0.391	0.208	0.921	0.955	0.961	2.592
ShiftMDModel	Conditioned full	Three-seed mean	0.402	0.206	0.919	0.957	0.965	2.591
ShiftMDModel	Conditioned full	Sample SD	0.001	0.001	0.002	0.004	0.004	0.002

Table S12: **RefDbSmallSet** protected partition (45 proteins): exact-assignment backbone RMSE (ppm). The selected static **ShiftTfnModel** used **RefDbSmallTleapSet**; the fixed first-retained-frame and uniform-mean baselines and conditioned-full **ShiftMDModel** used **RefDbSmallGromacsSet**.

Model	Route	Statistic	H	HA	CA	CB	C	N
ShiftTfnModel	Selected static	Fixed	0.269	0.141	0.604	0.623	0.675	1.716
ShiftTfnModel	Frame 15 baseline	Fixed	0.332	0.169	0.746	0.773	0.794	1.979
ShiftTfnModel	Uniform mean baseline	Fixed	0.288	0.153	0.680	0.692	0.733	1.905
ShiftMDModel	Conditioned full	Three-seed mean	0.301	0.151	0.679	0.694	0.737	1.904
ShiftMDModel	Conditioned full	Sample SD	0.002	0.001	0.001	0.004	0.003	0.002

Table S13: **RefDbSmallSet** protected partition (45 proteins): exact-assignment backbone MAE (ppm). The selected static **ShiftTfnModel** used **RefDbSmallTleapSet**; the fixed first-retained-frame and uniform-mean baselines and conditioned-full **ShiftMDModel** used **RefDbSmallGromacsSet**.

4.1 Complete protected residue–backbone detail

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
ALA	H	184	45	0.357	0.426	0.353	0.378	0.004
ALA	HA	182	43	0.167	0.189	0.193	0.180	0.002
ALA	CA	174	42	0.600	0.689	0.667	0.670	0.002
ALA	CB	164	42	0.736	0.897	0.802	0.807	0.009
ALA	C	134	35	0.912	1.072	0.955	0.976	0.021
ALA	N	171	44	1.923	2.433	2.296	2.296	0.001
ARG	H	191	42	0.332	0.360	0.330	0.342	0.001
ARG	HA	187	40	0.181	0.227	0.197	0.207	0.000
ARG	CA	184	40	0.851	1.043	0.943	0.941	0.002
ARG	CB	172	40	0.790	0.938	0.761	0.761	0.005
ARG	C	140	33	0.819	1.081	1.011	1.014	0.009
ARG	N	169	41	2.250	2.329	2.445	2.445	0.001
ASN	H	119	42	0.378	0.466	0.439	0.441	0.001
ASN	HA	120	41	0.182	0.218	0.204	0.208	0.004
ASN	CA	114	39	0.861	0.957	0.902	0.915	0.012
ASN	CB	109	39	0.773	0.891	0.877	0.872	0.004
ASN	C	72	30	0.828	1.007	0.895	0.928	0.030
ASN	N	111	39	2.513	2.692	2.606	2.604	0.003
ASP	H	182	43	0.395	0.422	0.395	0.404	0.003
ASP	HA	177	41	0.146	0.186	0.157	0.163	0.001
ASP	CA	171	40	0.918	1.083	1.027	1.020	0.008
ASP	CB	162	40	1.157	1.260	1.243	1.242	0.004
ASP	C	123	33	0.741	0.904	0.853	0.862	0.008
ASP	N	166	42	2.190	2.460	2.355	2.355	0.001
CYS	H	27	10	0.471	0.695	0.575	0.633	0.010
CYS	HA	27	10	0.252	0.294	0.279	0.255	0.016
CYS	CA	27	10	1.645	1.796	1.774	1.776	0.015
CYS	CB	26	10	2.779	2.811	2.736	2.711	0.021
CYS	C	15	6	1.353	1.407	1.399	1.383	0.027
CYS	N	24	9	2.914	3.424	3.431	3.426	0.008
GLN	H	181	42	0.336	0.423	0.369	0.376	0.002
GLN	HA	172	40	0.236	0.257	0.227	0.236	0.004
GLN	CA	170	39	0.707	0.857	0.818	0.816	0.002
GLN	CB	166	39	0.759	0.946	0.858	0.854	0.004
GLN	C	131	32	0.842	1.020	0.951	0.934	0.016
GLN	N	165	41	2.110	2.537	2.469	2.470	0.002
GLU	H	232	45	0.420	0.481	0.413	0.432	0.004
GLU	HA	222	43	0.189	0.226	0.194	0.194	0.004
GLU	CA	207	42	0.734	0.863	0.763	0.768	0.004
GLU	CB	196	41	0.759	1.010	0.908	0.922	0.013
GLU	C	165	35	0.882	0.987	0.967	0.965	0.003
GLU	N	215	44	2.069	2.502	2.172	2.173	0.002
GLY	H	260	45	0.380	0.447	0.368	0.386	0.002
GLY	HA	150	30	0.211	0.193	0.196	0.200	0.002
GLY	CA	238	40	0.599	0.729	0.676	0.681	0.008
GLY	C	177	32	1.066	1.089	1.070	1.057	0.035
GLY	N	225	41	3.244	3.305	3.193	3.190	0.005

continued

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
HIS	H	57	29	0.443	0.371	0.388	0.381	0.011
HIS	HA	59	27	0.301	0.272	0.239	0.261	0.005
HIS	CA	58	27	1.057	1.142	1.082	1.077	0.005
HIS	CB	55	26	0.844	0.940	0.888	0.894	0.005
HIS	C	39	20	0.791	0.788	0.829	0.818	0.013
HIS	N	53	28	3.161	3.607	3.517	3.518	0.002
ILE	H	179	44	0.363	0.525	0.474	0.486	0.003
ILE	HA	165	42	0.195	0.263	0.258	0.248	0.007
ILE	CA	157	41	0.823	0.990	1.049	0.998	0.045
ILE	CB	153	41	0.958	1.181	1.150	1.133	0.015
ILE	C	116	34	0.862	0.994	0.977	0.976	0.004
ILE	N	163	43	2.429	2.670	2.562	2.562	0.001
LEU	H	327	45	0.380	0.458	0.417	0.420	0.001
LEU	HA	308	43	0.172	0.246	0.220	0.210	0.004
LEU	CA	296	42	0.690	0.855	0.732	0.746	0.012
LEU	CB	285	42	0.873	1.199	1.052	1.061	0.011
LEU	C	225	35	0.878	0.992	0.918	0.926	0.007
LEU	N	302	44	2.142	2.453	2.341	2.340	0.001
LYS	H	254	45	0.308	0.367	0.329	0.341	0.003
LYS	HA	237	43	0.183	0.240	0.203	0.207	0.003
LYS	CA	216	42	0.863	1.069	0.947	0.946	0.001
LYS	CB	200	42	0.697	0.949	0.773	0.787	0.014
LYS	C	163	35	0.765	0.949	0.860	0.840	0.017
LYS	N	221	44	2.149	2.264	2.202	2.201	0.002
MET	H	46	26	0.349	0.356	0.338	0.334	0.007
MET	HA	54	30	0.237	0.273	0.236	0.238	0.009
MET	CA	50	28	0.822	1.251	0.998	1.003	0.017
MET	CB	44	28	1.470	1.611	1.389	1.398	0.012
MET	C	35	22	1.127	1.168	1.091	1.092	0.014
MET	N	41	24	1.891	2.093	1.851	1.853	0.003
PHE	H	105	43	0.353	0.472	0.396	0.394	0.009
PHE	HA	103	41	0.220	0.288	0.272	0.263	0.001
PHE	CA	97	40	0.919	1.171	1.187	1.184	0.008
PHE	CB	93	40	0.889	1.171	1.044	1.045	0.001
PHE	C	81	34	0.944	1.125	1.065	1.066	0.002
PHE	N	99	42	1.827	2.036	1.839	1.836	0.005
PRO	HA	158	43	0.186	0.225	0.213	0.200	0.006
PRO	CA	139	42	0.598	0.671	0.584	0.587	0.017
PRO	CB	135	42	0.757	0.775	0.733	0.734	0.004
PRO	C	98	34	0.993	1.050	0.990	1.003	0.013
SER	H	198	45	0.384	0.446	0.435	0.425	0.002
SER	HA	223	43	0.175	0.186	0.164	0.167	0.003
SER	CA	190	41	0.914	1.163	1.066	1.074	0.007
SER	CB	182	41	0.680	0.711	0.664	0.661	0.003
SER	C	143	33	0.911	1.067	0.949	0.966	0.016
SER	N	157	43	2.248	2.658	2.513	2.513	0.000
THR	H	172	43	0.356	0.439	0.381	0.394	0.002
THR	HA	166	41	0.189	0.244	0.210	0.203	0.002
THR	CA	166	40	0.675	0.971	1.006	0.976	0.028
THR	CB	160	40	0.767	0.889	0.762	0.774	0.015
THR	C	116	32	0.799	0.958	0.906	0.910	0.011

continued

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
THR	N	157	42	3.367	4.005	3.997	3.996	0.001
TRP	H	50	28	0.464	0.489	0.415	0.447	0.011
TRP	HA	50	28	0.199	0.247	0.196	0.203	0.003
TRP	CA	48	28	1.179	1.490	1.296	1.337	0.039
TRP	CB	48	28	0.912	1.192	1.105	1.100	0.005
TRP	C	37	25	1.210	1.395	1.231	1.280	0.042
TRP	N	47	28	2.142	2.043	2.367	2.364	0.005
TYR	H	92	44	0.362	0.465	0.396	0.412	0.001
TYR	HA	91	41	0.214	0.203	0.195	0.180	0.002
TYR	CA	91	41	0.945	1.146	1.175	1.149	0.024
TYR	CB	88	41	0.909	1.196	0.907	0.934	0.025
TYR	C	70	33	0.762	0.961	0.845	0.853	0.009
TYR	N	87	42	1.755	2.410	2.044	2.040	0.007
VAL	H	182	44	0.324	0.414	0.347	0.364	0.003
VAL	HA	174	42	0.237	0.216	0.190	0.184	0.001
VAL	CA	169	41	0.932	1.153	0.953	0.960	0.009
VAL	CB	160	41	0.837	0.942	0.829	0.815	0.012
VAL	C	127	34	0.849	1.055	0.952	0.962	0.009
VAL	N	165	42	2.204	2.880	2.656	2.651	0.008

Table S14: Protected residue–backbone RMSE (ppm), derived from the current exact-prediction ledger for the 45 protected **RefDbSmallSet** proteins. **ShiftMDModel** values are the mean and sample SD of the three independently fitted conditioned-full models.

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
ALA	H	184	45	0.272	0.339	0.273	0.293	0.004
ALA	HA	182	43	0.125	0.138	0.142	0.132	0.003
ALA	CA	174	42	0.464	0.530	0.515	0.515	0.001
ALA	CB	164	42	0.516	0.653	0.586	0.588	0.010
ALA	C	134	35	0.709	0.799	0.731	0.742	0.014
ALA	N	171	44	1.462	1.851	1.720	1.719	0.001
ARG	H	191	42	0.248	0.277	0.252	0.267	0.005
ARG	HA	187	40	0.131	0.169	0.141	0.147	0.002
ARG	CA	184	40	0.632	0.786	0.688	0.684	0.004
ARG	CB	172	40	0.577	0.701	0.594	0.588	0.006
ARG	C	140	33	0.653	0.806	0.766	0.763	0.006
ARG	N	169	41	1.658	1.817	1.906	1.906	0.001
ASN	H	119	42	0.244	0.312	0.274	0.302	0.005
ASN	HA	120	41	0.136	0.156	0.148	0.150	0.002
ASN	CA	114	39	0.690	0.767	0.689	0.705	0.015
ASN	CB	109	39	0.577	0.682	0.686	0.678	0.007
ASN	C	72	30	0.651	0.788	0.691	0.723	0.028
ASN	N	111	39	1.937	2.130	1.997	1.996	0.002
ASP	H	182	43	0.273	0.316	0.290	0.300	0.002
ASP	HA	177	41	0.114	0.138	0.126	0.127	0.002
ASP	CA	171	40	0.674	0.819	0.777	0.769	0.008
ASP	CB	162	40	0.692	0.789	0.757	0.757	0.006
ASP	C	123	33	0.565	0.708	0.627	0.638	0.012
ASP	N	166	42	1.745	1.959	1.876	1.877	0.002
CYS	H	27	10	0.373	0.525	0.446	0.472	0.009
CYS	HA	27	10	0.215	0.228	0.209	0.199	0.012
CYS	CA	27	10	1.192	1.464	1.396	1.413	0.016
CYS	CB	26	10	1.941	2.079	2.003	1.996	0.010
CYS	C	15	6	1.116	1.137	1.078	1.074	0.039
CYS	N	24	9	2.022	2.466	2.183	2.184	0.002
GLN	H	181	42	0.249	0.325	0.269	0.284	0.003
GLN	HA	172	40	0.165	0.183	0.172	0.173	0.003
GLN	CA	170	39	0.577	0.692	0.648	0.646	0.005
GLN	CB	166	39	0.604	0.755	0.685	0.684	0.001
GLN	C	131	32	0.668	0.834	0.771	0.752	0.017
GLN	N	165	41	1.675	1.994	1.943	1.945	0.004
GLU	H	232	45	0.298	0.357	0.301	0.313	0.003
GLU	HA	222	43	0.135	0.168	0.142	0.144	0.003
GLU	CA	207	42	0.605	0.668	0.592	0.598	0.005
GLU	CB	196	41	0.577	0.744	0.679	0.689	0.009
GLU	C	165	35	0.693	0.805	0.768	0.754	0.012
GLU	N	215	44	1.524	1.865	1.728	1.728	0.000
GLY	H	260	45	0.269	0.307	0.254	0.273	0.001
GLY	HA	150	30	0.128	0.136	0.146	0.149	0.004
GLY	CA	238	40	0.423	0.523	0.471	0.479	0.008
GLY	C	177	32	0.646	0.749	0.698	0.709	0.014
GLY	N	225	41	1.876	2.037	1.963	1.959	0.007
HIS	H	57	29	0.319	0.281	0.299	0.293	0.011
HIS	HA	59	27	0.188	0.206	0.177	0.193	0.002
HIS	CA	58	27	0.826	0.900	0.863	0.845	0.017

continued

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
HIS	CB	55	26	0.718	0.796	0.715	0.734	0.016
HIS	C	39	20	0.651	0.642	0.611	0.617	0.013
HIS	N	53	28	2.367	2.892	2.674	2.677	0.004
ILE	H	179	44	0.279	0.393	0.356	0.370	0.003
ILE	HA	165	42	0.148	0.207	0.197	0.187	0.004
ILE	CA	157	41	0.683	0.785	0.807	0.775	0.028
ILE	CB	153	41	0.749	0.932	0.884	0.865	0.016
ILE	C	116	34	0.657	0.790	0.781	0.775	0.005
ILE	N	163	43	1.886	2.059	2.011	2.011	0.000
LEU	H	327	45	0.286	0.360	0.320	0.321	0.003
LEU	HA	308	43	0.130	0.181	0.156	0.151	0.002
LEU	CA	296	42	0.497	0.664	0.549	0.563	0.013
LEU	CB	285	42	0.660	0.949	0.805	0.818	0.013
LEU	C	225	35	0.684	0.788	0.709	0.725	0.014
LEU	N	302	44	1.589	1.866	1.796	1.795	0.001
LYS	H	254	45	0.227	0.290	0.258	0.269	0.001
LYS	HA	237	43	0.138	0.173	0.147	0.147	0.002
LYS	CA	216	42	0.602	0.779	0.689	0.684	0.005
LYS	CB	200	42	0.493	0.702	0.595	0.601	0.007
LYS	C	163	35	0.618	0.730	0.683	0.663	0.018
LYS	N	221	44	1.674	1.768	1.721	1.719	0.003
MET	H	46	26	0.282	0.289	0.271	0.270	0.006
MET	HA	54	30	0.178	0.202	0.186	0.191	0.004
MET	CA	50	28	0.649	1.019	0.823	0.816	0.029
MET	CB	44	28	0.868	1.213	0.984	1.006	0.022
MET	C	35	22	0.855	0.934	0.880	0.880	0.016
MET	N	41	24	1.485	1.670	1.453	1.460	0.012
PHE	H	105	43	0.263	0.368	0.312	0.308	0.010
PHE	HA	103	41	0.178	0.230	0.218	0.209	0.002
PHE	CA	97	40	0.722	0.937	0.901	0.914	0.012
PHE	CB	93	40	0.674	0.878	0.815	0.808	0.007
PHE	C	81	34	0.792	0.919	0.884	0.882	0.002
PHE	N	99	42	1.441	1.598	1.355	1.352	0.005
PRO	HA	158	43	0.112	0.154	0.138	0.133	0.002
PRO	CA	139	42	0.471	0.529	0.474	0.465	0.012
PRO	CB	135	42	0.550	0.548	0.530	0.526	0.004
PRO	C	98	34	0.793	0.796	0.790	0.786	0.004
SER	H	198	45	0.274	0.330	0.302	0.298	0.002
SER	HA	223	43	0.126	0.130	0.114	0.117	0.004
SER	CA	190	41	0.697	0.850	0.764	0.768	0.008
SER	CB	182	41	0.500	0.530	0.494	0.491	0.003
SER	C	143	33	0.690	0.806	0.702	0.720	0.016
SER	N	157	43	1.709	1.823	1.763	1.763	0.001
THR	H	172	43	0.269	0.335	0.285	0.296	0.004
THR	HA	166	41	0.148	0.189	0.163	0.158	0.002
THR	CA	166	40	0.504	0.744	0.756	0.730	0.024
THR	CB	160	40	0.514	0.661	0.549	0.555	0.009
THR	C	116	32	0.574	0.698	0.642	0.656	0.018
THR	N	157	42	2.276	2.749	2.839	2.838	0.003
TRP	H	50	28	0.360	0.384	0.320	0.363	0.014
TRP	HA	50	28	0.154	0.190	0.154	0.153	0.003

continued

Residue	Atom	Assignments	Proteins	Static ShiftTfnModel	First-retained-frame ShiftTfnModel	Uniform parent mean	ShiftMDModel mean	ShiftMDModel SD
TRP	CA	48	28	0.845	1.147	1.016	1.055	0.037
TRP	CB	48	28	0.724	0.952	0.906	0.896	0.008
TRP	C	37	25	1.030	1.130	1.090	1.118	0.025
TRP	N	47	28	1.644	1.637	1.901	1.898	0.006
TYR	H	92	44	0.264	0.365	0.305	0.321	0.005
TYR	HA	91	41	0.147	0.155	0.150	0.137	0.002
TYR	CA	91	41	0.690	0.887	0.874	0.859	0.012
TYR	CB	88	41	0.696	0.914	0.693	0.721	0.026
TYR	C	70	33	0.612	0.774	0.653	0.657	0.004
TYR	N	87	42	1.345	1.824	1.583	1.580	0.005
VAL	H	182	44	0.240	0.318	0.259	0.284	0.002
VAL	HA	174	42	0.183	0.173	0.153	0.146	0.001
VAL	CA	169	41	0.717	0.884	0.728	0.724	0.006
VAL	CB	160	41	0.688	0.749	0.673	0.662	0.011
VAL	C	127	34	0.627	0.841	0.720	0.735	0.014
VAL	N	165	42	1.718	2.200	2.087	2.083	0.007

Table S15: Protected residue–backbone MAE (ppm), derived from the current exact-prediction ledger for the 45 protected **RefDbSmallSet** proteins. **ShiftMDModel** values are the mean and sample SD of the three independently fitted conditioned-full models.

5 Static External Results

All conditions below are independently refitted static **ShiftTfnModel** source-contrast conditions trained on all 173 **RefDbSmallTleapSet** proteins and then frozen. The condition labels Plain, ORCA transfer, MOPAC and AIMNet2 belong to this separate source contrast; they are not aliases for the four canonical development forms. No first-retained-frame **ShiftTfnModel** or **ShiftMDModel** was applied to either static external set.

Condition	Metric	H	HA	CA	CB	C	N	Role-balanced loss
RefDbTestSmallSet (386 proteins)								
Plain	RMSE	0.450	0.261	0.927	1.139	1.423	2.562	0.492
Plain	MAE	0.323	0.184	0.659	0.710	0.774	1.857	—
ORCA transfer	RMSE	0.391	0.223	0.861	1.059	1.357	2.257	0.416
ORCA transfer	MAE	0.284	0.157	0.602	0.644	0.692	1.607	—
MOPAC	RMSE	0.418	0.236	0.913	1.116	1.394	2.590	0.443
MOPAC	MAE	0.305	0.168	0.647	0.690	0.742	1.887	—
AIMNet2	RMSE	0.438	0.255	0.962	1.157	1.438	2.611	0.479
AIMNet2	MAE	0.318	0.184	0.688	0.732	0.799	1.912	—
RefDbTestLargeSet (1,746 proteins)								
Plain	RMSE	0.486	0.292	1.039	1.363	1.518	2.857	0.578
Plain	MAE	0.346	0.196	0.720	0.784	0.864	1.994	—
ORCA transfer	RMSE	0.429	0.260	0.965	1.287	1.463	2.530	0.512
ORCA transfer	MAE	0.307	0.169	0.649	0.711	0.784	1.711	—
MOPAC	RMSE	0.458	0.272	1.021	1.348	1.500	2.866	0.541
MOPAC	MAE	0.331	0.180	0.701	0.769	0.839	2.010	—
AIMNet2	RMSE	0.473	0.288	1.064	1.380	1.531	2.881	0.570
AIMNet2	MAE	0.341	0.196	0.742	0.802	0.885	2.031	—

Table S16: Current static-source comparison on the nested **RefDbTestSmallSet** and **RefDbTestLargeSet**. Backbone RMSE and MAE are in ppm; the dimensionless role-balanced loss is printed only on RMSE rows.

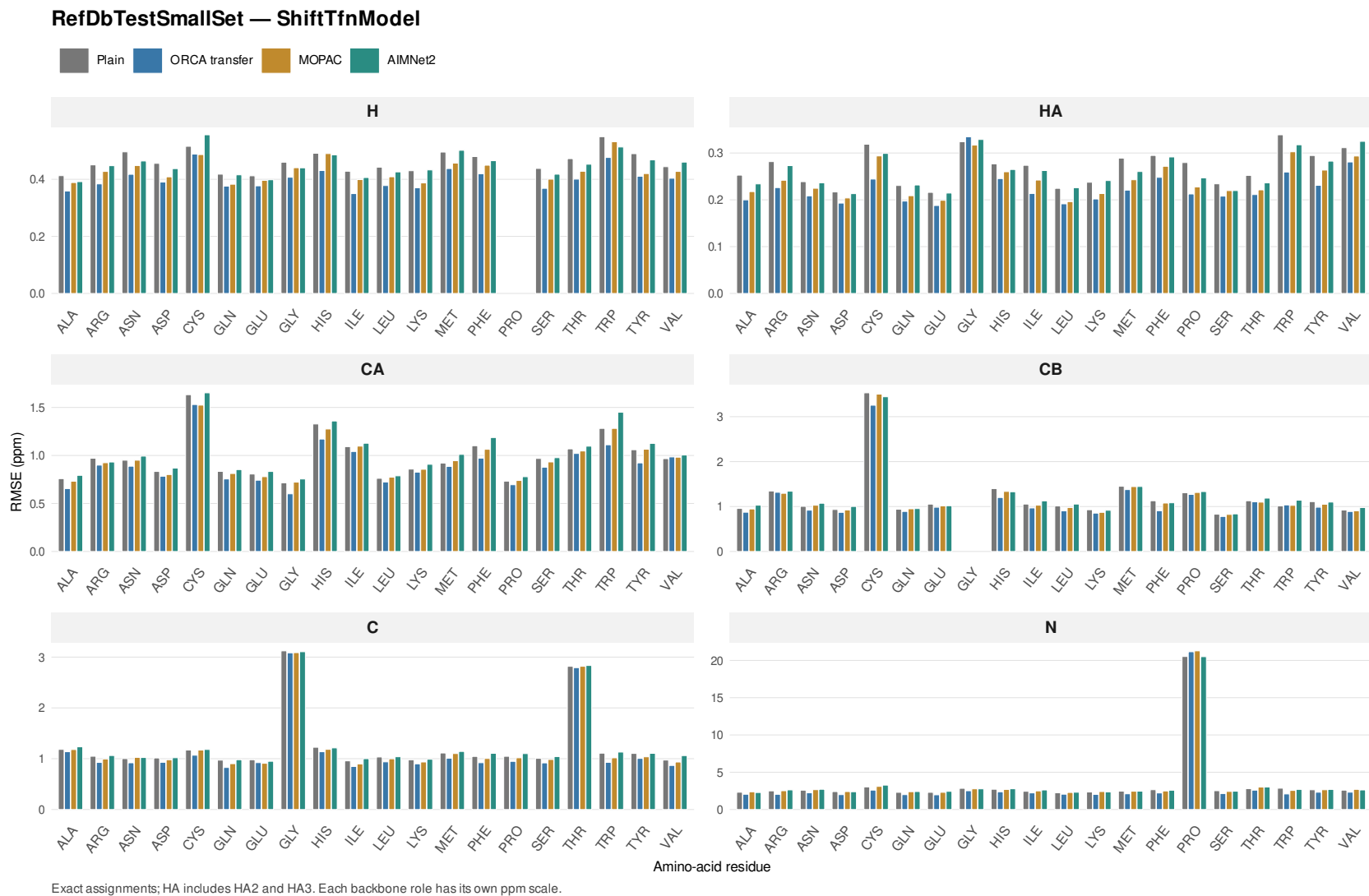


Figure S1: Residue-backbone RMSE (ppm) for the four static **ShiftTfnModel** source-contrast conditions on **RefDbTestSmallSet**.

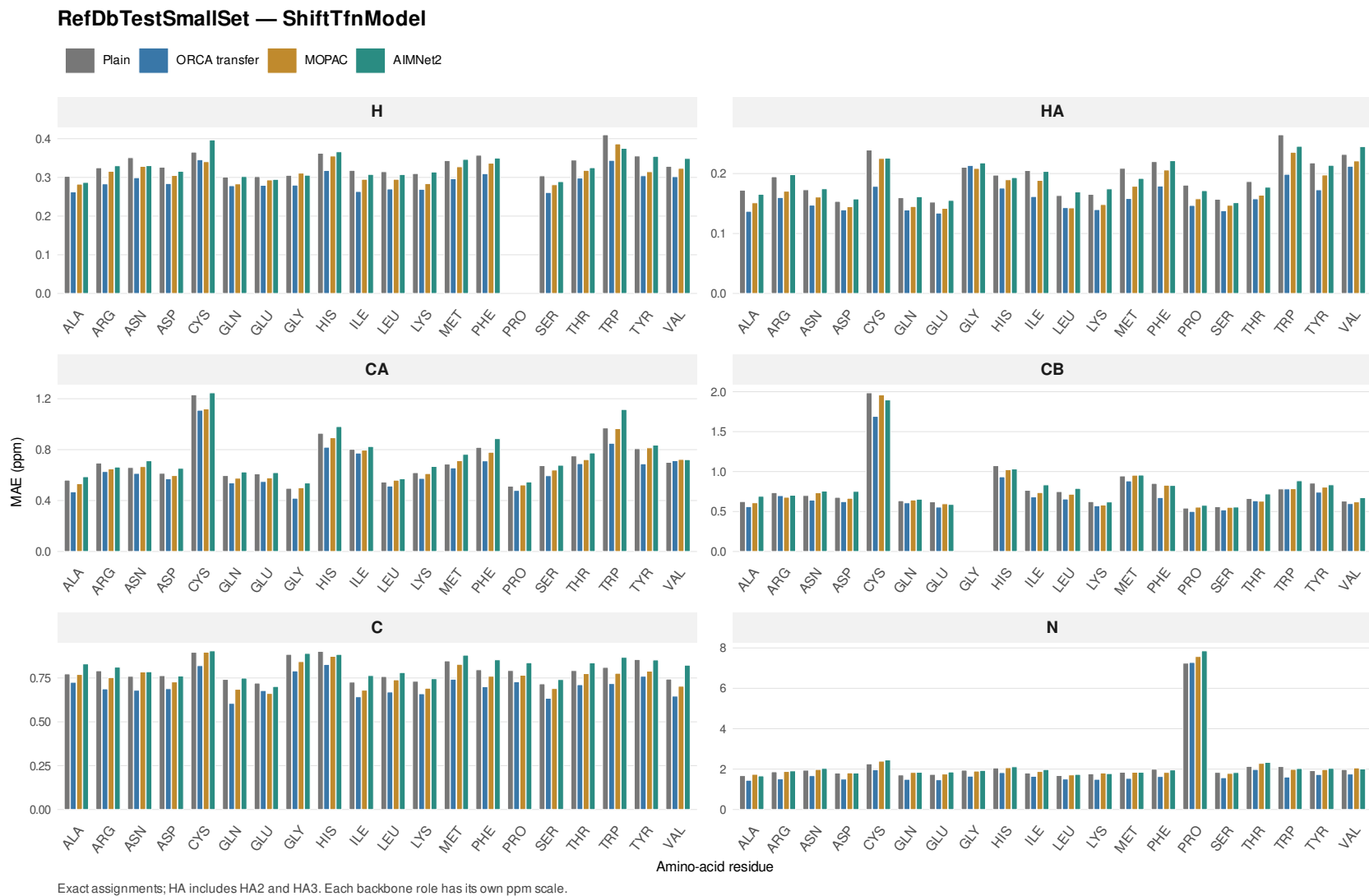


Figure S2: Residue–backbone MAE (ppm) for the four static **ShiftTfnModel** source-contrast conditions on **RefDbTestSmallSet**.

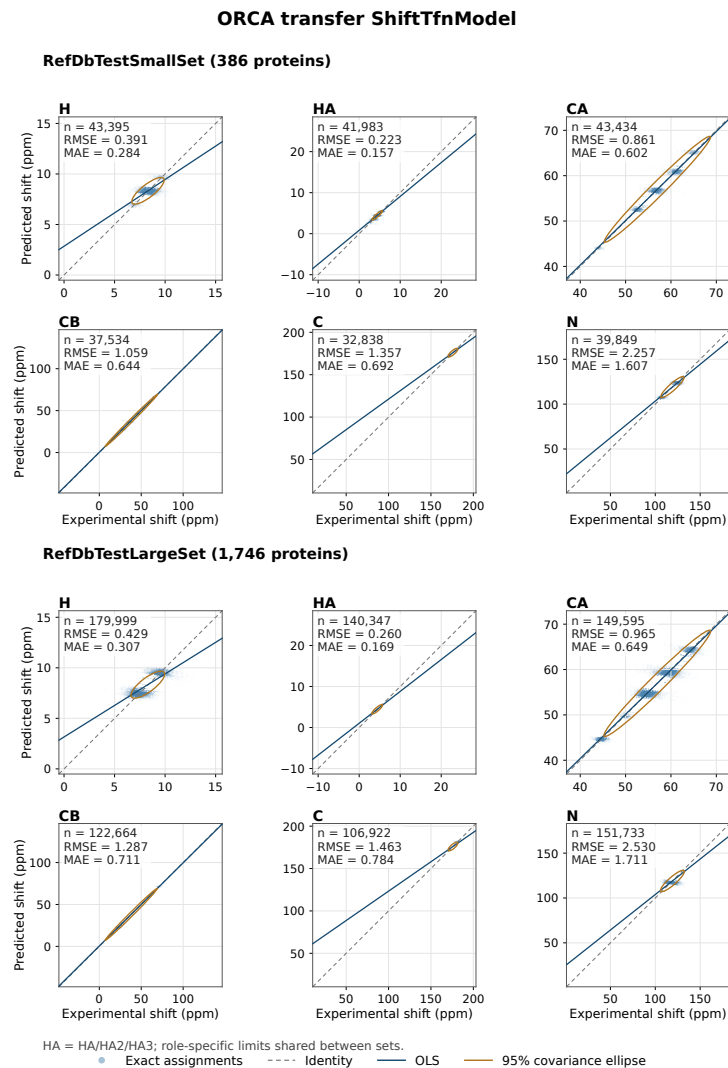


Figure S3: Predicted and experimental backbone shifts for the ORCA-transfer static **ShiftTfnModel** all-173 refit on **RefDbTestSmallSet** and **RefDbTestLargeSet**. Solid lines are ordinary least squares; dashed lines mark exact agreement; ovals are 95% covariance data ellipses.

5.1 Complete RefDbTestSmallSet residue-backbone detail

Residue	Atom	Assignments	Proteins	Plain	ORCA	transfer	MOPAC	AIMNet2
ALA	H	3,295	374	0.413		0.359	0.389	0.392
ALA	HA	3,098	359	0.253		0.200	0.218	0.234
ALA	CA	3,192	358	0.758		0.655	0.732	0.793
ALA	CB	2,968	351	0.957		0.873	0.946	1.029
ALA	C	2,375	293	1.183		1.137	1.180	1.233
ALA	N	3,072	362	2.310		2.045	2.373	2.278
ARG	H	2,150	373	0.450		0.384	0.427	0.448
ARG	HA	2,003	358	0.282		0.226	0.242	0.273
ARG	CA	2,071	356	0.971		0.899	0.924	0.932
ARG	CB	1,893	347	1.346		1.317	1.295	1.341
ARG	C	1,580	289	1.048		0.926	0.995	1.062
ARG	N	1,960	361	2.497		2.038	2.514	2.627
ASN	H	1,787	372	0.496		0.418	0.448	0.464
ASN	HA	1,686	356	0.239		0.209	0.225	0.236
ASN	CA	1,712	354	0.950		0.888	0.950	0.993
ASN	CB	1,608	344	1.004		0.919	1.029	1.073
ASN	C	1,268	287	1.001		0.919	1.026	1.023
ASN	N	1,657	358	2.590		2.242	2.655	2.710
ASP	H	2,549	379	0.456		0.390	0.409	0.437
ASP	HA	2,405	364	0.217		0.193	0.204	0.213
ASP	CA	2,459	362	0.833		0.784	0.801	0.868
ASP	CB	2,292	354	0.934		0.870	0.924	1.002
ASP	C	1,855	295	1.014		0.929	0.979	1.019
ASP	N	2,388	366	2.400		1.990	2.403	2.382
CYS	H	750	294	0.516		0.488	0.487	0.556
CYS	HA	710	284	0.319		0.244	0.294	0.299
CYS	CA	673	280	1.633		1.530	1.524	1.652
CYS	CB	610	268	3.530		3.255	3.502	3.443
CYS	C	500	222	1.171		1.068	1.170	1.181
CYS	N	643	279	3.029		2.600	3.097	3.269
GLN	H	1,940	364	0.418		0.376	0.383	0.416
GLN	HA	1,874	349	0.231		0.197	0.209	0.232
GLN	CA	1,854	349	0.834		0.755	0.812	0.853
GLN	CB	1,730	341	0.939		0.891	0.949	0.954
GLN	C	1,381	290	0.971		0.827	0.902	0.980
GLN	N	1,769	353	2.299		1.991	2.386	2.407
GLU	H	3,583	375	0.412		0.377	0.396	0.398
GLU	HA	3,360	361	0.216		0.188	0.199	0.214
GLU	CA	3,444	361	0.806		0.742	0.779	0.833
GLU	CB	3,177	354	1.054		0.985	1.015	1.017
GLU	C	2,678	296	0.979		0.923	0.909	0.949
GLU	N	3,322	366	2.299		1.968	2.303	2.435
GLY	H	3,207	380	0.460		0.407	0.440	0.440
GLY	HA	1,880	260	0.324		0.335	0.317	0.329
GLY	CA	2,990	361	0.715		0.600	0.722	0.755
GLY	C	2,217	294	3.123		3.084	3.087	3.105

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
GLY	N	2,805	364	2.838	2.512	2.770	2.767
HIS	H	925	323	0.491	0.431	0.490	0.486
HIS	HA	894	313	0.277	0.245	0.260	0.265
HIS	CA	910	313	1.327	1.171	1.275	1.359
HIS	CB	866	309	1.399	1.201	1.336	1.329
HIS	C	702	251	1.225	1.137	1.184	1.212
HIS	N	832	308	2.715	2.376	2.692	2.777
ILE	H	2,409	374	0.428	0.350	0.399	0.406
ILE	HA	2,256	360	0.274	0.214	0.242	0.263
ILE	CA	2,289	356	1.092	1.041	1.097	1.127
ILE	CB	2,124	350	1.055	0.968	1.030	1.126
ILE	C	1,767	293	0.957	0.847	0.897	1.000
ILE	N	2,244	361	2.440	2.224	2.495	2.619
LEU	H	4,153	378	0.442	0.378	0.409	0.426
LEU	HA	3,887	364	0.225	0.191	0.196	0.226
LEU	CA	3,961	358	0.762	0.724	0.773	0.788
LEU	CB	3,712	352	1.012	0.903	0.978	1.055
LEU	C	3,026	294	1.033	0.938	0.998	1.038
LEU	N	3,846	365	2.243	2.052	2.284	2.312
LYS	H	3,184	374	0.430	0.370	0.387	0.433
LYS	HA	3,006	359	0.237	0.202	0.213	0.241
LYS	CA	3,055	358	0.858	0.826	0.856	0.908
LYS	CB	2,818	351	0.928	0.854	0.871	0.917
LYS	C	2,334	295	0.978	0.899	0.935	0.989
LYS	N	2,971	363	2.344	2.029	2.404	2.363
MET	H	896	330	0.495	0.438	0.457	0.502
MET	HA	896	316	0.289	0.221	0.243	0.260
MET	CA	925	319	0.921	0.886	0.945	1.011
MET	CB	868	311	1.451	1.377	1.444	1.446
MET	C	698	261	1.112	1.010	1.101	1.143
MET	N	852	320	2.465	2.106	2.446	2.458
PHE	H	1,621	360	0.480	0.419	0.449	0.465
PHE	HA	1,545	346	0.295	0.248	0.272	0.292
PHE	CA	1,539	345	1.101	0.972	1.064	1.187
PHE	CB	1,447	335	1.129	0.907	1.077	1.083
PHE	C	1,174	287	1.045	0.921	1.004	1.106
PHE	N	1,497	350	2.651	2.217	2.488	2.584
PRO	HA	1,998	352	0.280	0.213	0.227	0.247
PRO	CA	1,973	353	0.732	0.695	0.739	0.779
PRO	CB	1,814	344	1.307	1.270	1.313	1.333
PRO	C	1,382	281	1.047	0.948	1.018	1.100
PRO	N	29	7	20.520	21.157	21.278	20.501
SER	H	3,114	376	0.438	0.368	0.400	0.418
SER	HA	3,044	362	0.234	0.208	0.220	0.220
SER	CA	2,936	359	0.968	0.878	0.932	0.977
SER	CB	2,669	351	0.829	0.779	0.824	0.837
SER	C	2,272	295	1.006	0.917	0.984	1.042
SER	N	2,717	362	2.516	2.140	2.421	2.451
THR	H	2,538	371	0.472	0.401	0.428	0.453
THR	HA	2,417	356	0.252	0.211	0.221	0.236

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
THR	CA	2,400	355	1.068	1.020	1.046	1.096
THR	CB	2,225	348	1.128	1.105	1.098	1.187
THR	C	1,754	290	2.818	2.792	2.819	2.835
THR	N	2,323	359	2.797	2.595	3.008	3.018
TRP	H	590	274	0.549	0.476	0.532	0.513
TRP	HA	569	264	0.339	0.259	0.303	0.318
TRP	CA	560	262	1.282	1.111	1.282	1.450
TRP	CB	520	251	1.013	1.036	1.026	1.141
TRP	C	439	212	1.109	0.928	1.017	1.130
TRP	N	540	258	2.853	2.086	2.555	2.697
TYR	H	1,363	366	0.490	0.411	0.420	0.468
TYR	HA	1,302	351	0.295	0.231	0.264	0.283
TYR	CA	1,302	349	1.059	0.922	1.065	1.125
TYR	CB	1,226	339	1.109	0.985	1.051	1.099
TYR	C	992	285	1.104	1.006	1.037	1.105
TYR	N	1,264	349	2.641	2.315	2.644	2.691
VAL	H	3,341	377	0.444	0.404	0.428	0.460
VAL	HA	3,153	362	0.311	0.281	0.294	0.325
VAL	CA	3,189	359	0.967	0.984	0.982	1.004
VAL	CB	2,967	353	0.922	0.890	0.905	0.977
VAL	C	2,444	297	0.973	0.867	0.934	1.061
VAL	N	3,118	365	2.594	2.321	2.700	2.619

Table S17: Complete residue–backbone RMSE (ppm) for the four independently refitted static **ShiftTfmModel** source-contrast conditions on **RefDbTestSmallSet**. No first-retained-frame or trajectory model is included.

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
ALA	H	3,295	374	0.303	0.263	0.283	0.287
ALA	HA	3,098	359	0.172	0.137	0.151	0.165
ALA	CA	3,192	358	0.560	0.469	0.532	0.587
ALA	CB	2,968	351	0.625	0.562	0.609	0.691
ALA	C	2,375	293	0.773	0.725	0.770	0.830
ALA	N	3,072	362	1.676	1.448	1.741	1.662
ARG	H	2,150	373	0.325	0.283	0.316	0.330
ARG	HA	2,003	358	0.195	0.160	0.171	0.198
ARG	CA	2,071	356	0.695	0.628	0.649	0.663
ARG	CB	1,893	347	0.737	0.699	0.679	0.703
ARG	C	1,580	289	0.791	0.688	0.752	0.812
ARG	N	1,960	361	1.865	1.515	1.882	1.912
ASN	H	1,787	372	0.351	0.299	0.328	0.330
ASN	HA	1,686	356	0.173	0.147	0.161	0.175
ASN	CA	1,712	354	0.660	0.613	0.666	0.712
ASN	CB	1,608	344	0.701	0.643	0.735	0.756
ASN	C	1,268	287	0.760	0.680	0.784	0.786
ASN	N	1,657	358	1.947	1.674	1.981	2.035
ASP	H	2,549	379	0.326	0.284	0.305	0.316
ASP	HA	2,405	364	0.154	0.139	0.145	0.158
ASP	CA	2,459	362	0.614	0.571	0.596	0.653
ASP	CB	2,292	354	0.677	0.624	0.666	0.752
ASP	C	1,855	295	0.763	0.689	0.727	0.761
ASP	N	2,388	366	1.806	1.507	1.814	1.807
CYS	H	750	294	0.365	0.346	0.341	0.397
CYS	HA	710	284	0.239	0.179	0.225	0.226
CYS	CA	673	280	1.231	1.109	1.120	1.246
CYS	CB	610	268	1.985	1.691	1.960	1.897
CYS	C	500	222	0.896	0.820	0.897	0.905
CYS	N	643	279	2.261	1.971	2.393	2.455
GLN	H	1,940	364	0.301	0.278	0.283	0.302
GLN	HA	1,874	349	0.160	0.139	0.145	0.161
GLN	CA	1,854	349	0.596	0.539	0.576	0.625
GLN	CB	1,730	341	0.635	0.609	0.644	0.654
GLN	C	1,381	290	0.742	0.605	0.685	0.749
GLN	N	1,769	353	1.710	1.493	1.835	1.841
GLU	H	3,583	375	0.302	0.280	0.293	0.295
GLU	HA	3,360	361	0.152	0.134	0.142	0.155
GLU	CA	3,444	361	0.610	0.550	0.578	0.619
GLU	CB	3,177	354	0.621	0.556	0.597	0.589
GLU	C	2,678	296	0.721	0.678	0.662	0.701
GLU	N	3,322	366	1.738	1.472	1.757	1.857
GLY	H	3,207	380	0.305	0.280	0.311	0.306
GLY	HA	1,880	260	0.210	0.213	0.209	0.218
GLY	CA	2,990	361	0.496	0.418	0.501	0.538
GLY	C	2,217	294	0.885	0.790	0.843	0.890
GLY	N	2,805	364	1.947	1.651	1.901	1.933
HIS	H	925	323	0.362	0.318	0.355	0.366
HIS	HA	894	313	0.197	0.176	0.190	0.193

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
HIS	CA	910	313	0.928	0.819	0.893	0.981
HIS	CB	866	309	1.074	0.934	1.024	1.032
HIS	C	702	251	0.902	0.827	0.873	0.884
HIS	N	832	308	2.057	1.826	2.069	2.115
ILE	H	2,409	374	0.318	0.264	0.295	0.308
ILE	HA	2,256	360	0.205	0.162	0.189	0.204
ILE	CA	2,289	356	0.802	0.773	0.796	0.824
ILE	CB	2,124	350	0.765	0.683	0.737	0.833
ILE	C	1,767	293	0.727	0.643	0.680	0.764
ILE	N	2,244	361	1.807	1.643	1.881	1.977
LEU	H	4,153	378	0.315	0.270	0.295	0.307
LEU	HA	3,887	364	0.163	0.143	0.143	0.169
LEU	CA	3,961	358	0.546	0.514	0.560	0.571
LEU	CB	3,712	352	0.749	0.657	0.717	0.789
LEU	C	3,026	294	0.758	0.670	0.739	0.781
LEU	N	3,846	365	1.682	1.510	1.712	1.738
LYS	H	3,184	374	0.310	0.269	0.284	0.314
LYS	HA	3,006	359	0.165	0.140	0.148	0.174
LYS	CA	3,055	358	0.619	0.574	0.612	0.668
LYS	CB	2,818	351	0.624	0.572	0.582	0.620
LYS	C	2,334	295	0.731	0.660	0.692	0.746
LYS	N	2,971	363	1.768	1.495	1.808	1.777
MET	H	896	330	0.344	0.297	0.327	0.347
MET	HA	896	316	0.209	0.158	0.179	0.192
MET	CA	925	319	0.686	0.657	0.713	0.763
MET	CB	868	311	0.945	0.883	0.955	0.956
MET	C	698	261	0.847	0.743	0.828	0.880
MET	N	852	320	1.844	1.541	1.843	1.842
PHE	H	1,621	360	0.357	0.310	0.337	0.350
PHE	HA	1,545	346	0.220	0.179	0.206	0.221
PHE	CA	1,539	345	0.818	0.712	0.779	0.886
PHE	CB	1,447	335	0.852	0.675	0.829	0.826
PHE	C	1,174	287	0.797	0.700	0.760	0.853
PHE	N	1,497	350	1.996	1.633	1.836	1.959
PRO	HA	1,998	352	0.181	0.147	0.158	0.171
PRO	CA	1,973	353	0.513	0.479	0.522	0.546
PRO	CB	1,814	344	0.543	0.501	0.555	0.578
PRO	C	1,382	281	0.792	0.728	0.766	0.837
PRO	N	29	7	7.246	7.280	7.574	7.854
SER	H	3,114	376	0.304	0.261	0.281	0.289
SER	HA	3,044	362	0.157	0.138	0.147	0.151
SER	CA	2,936	359	0.674	0.595	0.640	0.677
SER	CB	2,669	351	0.560	0.521	0.553	0.556
SER	C	2,272	295	0.716	0.634	0.690	0.741
SER	N	2,717	362	1.837	1.574	1.787	1.830
THR	H	2,538	371	0.345	0.298	0.318	0.325
THR	HA	2,417	356	0.187	0.158	0.164	0.177
THR	CA	2,400	355	0.751	0.690	0.722	0.774
THR	CB	2,225	348	0.663	0.635	0.629	0.720
THR	C	1,754	290	0.792	0.712	0.775	0.836

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
THR	N	2,323	359	2.133	1.981	2.284	2.332
TRP	H	590	274	0.410	0.344	0.387	0.375
TRP	HA	569	264	0.265	0.199	0.236	0.245
TRP	CA	560	262	0.970	0.850	0.965	1.115
TRP	CB	520	251	0.783	0.783	0.785	0.885
TRP	C	439	212	0.810	0.718	0.777	0.868
TRP	N	540	258	2.125	1.608	1.985	2.029
TYR	H	1,363	366	0.355	0.304	0.315	0.354
TYR	HA	1,302	351	0.218	0.173	0.198	0.214
TYR	CA	1,302	349	0.807	0.689	0.814	0.836
TYR	CB	1,226	339	0.856	0.742	0.805	0.834
TYR	C	992	285	0.855	0.760	0.789	0.853
TYR	N	1,264	349	1.924	1.733	1.978	2.037
VAL	H	3,341	377	0.329	0.302	0.323	0.349
VAL	HA	3,153	362	0.232	0.212	0.221	0.245
VAL	CA	3,189	359	0.700	0.714	0.723	0.721
VAL	CB	2,967	353	0.631	0.599	0.621	0.674
VAL	C	2,444	297	0.744	0.647	0.703	0.823
VAL	N	3,118	365	1.978	1.756	2.059	2.006

Table S18: Complete residue–backbone MAE (ppm) for the four independently refitted static **ShiftTfnModel** source-contrast conditions on **RefDbTestSmallSet**. No first-retained-frame or trajectory model is included.

5.2 Complete RefDbTestLargeSet residue-backbone detail

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
ALA	H	14,007	1,656	0.445	0.392	0.414	0.428
ALA	HA	10,471	1,400	0.255	0.212	0.226	0.242
ALA	CA	11,291	1,227	0.843	0.730	0.801	0.841
ALA	CB	10,013	1,159	1.239	1.177	1.216	1.256
ALA	C	8,064	910	1.354	1.313	1.357	1.396
ALA	N	12,049	1,386	2.983	2.745	3.013	2.960
ARG	H	8,937	1,620	0.480	0.423	0.457	0.470
ARG	HA	6,885	1,373	0.275	0.239	0.252	0.284
ARG	CA	7,051	1,198	1.082	1.000	1.039	1.043
ARG	CB	6,121	1,128	1.254	1.180	1.189	1.213
ARG	C	5,059	889	1.190	1.092	1.141	1.207
ARG	N	7,423	1,350	2.688	2.254	2.733	2.853
ASN	H	7,836	1,630	0.521	0.442	0.473	0.486
ASN	HA	6,063	1,376	0.250	0.218	0.241	0.251
ASN	CA	6,183	1,193	1.023	0.945	1.011	1.061
ASN	CB	5,522	1,122	1.066	0.986	1.081	1.134
ASN	C	4,356	881	1.185	1.124	1.195	1.196
ASN	N	6,490	1,345	2.764	2.388	2.802	2.821
ASP	H	11,068	1,647	0.486	0.433	0.457	0.480
ASP	HA	8,361	1,388	0.224	0.199	0.209	0.223
ASP	CA	8,892	1,210	0.953	0.873	0.917	0.978
ASP	CB	7,931	1,144	1.549	1.505	1.553	1.596
ASP	C	6,354	899	1.180	1.111	1.171	1.176
ASP	N	9,553	1,370	2.752	2.299	2.794	2.704
CYS	H	4,332	1,214	0.606	0.534	0.553	0.576
CYS	HA	3,774	1,033	0.590	0.565	0.585	0.583
CYS	CA	2,412	845	1.751	1.678	1.693	1.770
CYS	CB	2,115	782	3.705	3.443	3.743	3.767
CYS	C	1,757	632	1.419	1.391	1.477	1.477
CYS	N	2,832	957	3.271	2.657	3.093	3.327
GLN	H	7,491	1,564	0.441	0.396	0.413	0.424
GLN	HA	5,879	1,315	0.242	0.211	0.221	0.246
GLN	CA	6,045	1,181	0.927	0.846	0.909	0.942
GLN	CB	5,396	1,115	1.143	1.068	1.133	1.168
GLN	C	4,299	878	1.054	0.955	1.019	1.056
GLN	N	6,356	1,321	2.435	2.077	2.479	2.530
GLU	H	14,014	1,654	0.441	0.396	0.419	0.426
GLU	HA	10,649	1,398	0.224	0.197	0.207	0.223
GLU	CA	11,489	1,236	0.938	0.870	0.920	0.954
GLU	CB	10,150	1,163	1.110	1.045	1.093	1.085
GLU	C	8,429	920	2.105	2.072	2.068	2.089
GLU	N	12,179	1,397	2.615	2.289	2.605	2.713
GLY	H	13,701	1,679	0.501	0.462	0.494	0.496
GLY	HA	6,916	887	0.410	0.412	0.402	0.408
GLY	CA	10,685	1,229	0.813	0.700	0.807	0.841
GLY	C	7,570	913	2.668	2.645	2.645	2.654

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
GLY	N	11,299	1,389	3.404	3.077	3.321	3.314
HIS	H	4,061	1,411	0.574	0.510	0.562	0.563
HIS	HA	3,139	1,175	0.341	0.309	0.322	0.326
HIS	CA	3,301	1,077	1.411	1.271	1.361	1.451
HIS	CB	2,926	1,002	1.695	1.538	1.656	1.642
HIS	C	2,332	789	1.319	1.200	1.287	1.290
HIS	N	3,315	1,168	3.027	2.578	2.929	3.008
ILE	H	9,937	1,628	0.470	0.397	0.449	0.445
ILE	HA	7,514	1,374	0.289	0.228	0.259	0.275
ILE	CA	8,121	1,204	1.166	1.097	1.164	1.213
ILE	CB	7,194	1,142	1.143	1.065	1.129	1.188
ILE	C	5,817	896	1.147	1.041	1.096	1.183
ILE	N	8,575	1,350	2.708	2.494	2.764	2.876
LEU	H	16,096	1,681	0.480	0.418	0.447	0.460
LEU	HA	12,081	1,418	0.243	0.212	0.216	0.243
LEU	CA	13,137	1,228	0.853	0.818	0.860	0.868
LEU	CB	11,629	1,155	1.143	1.041	1.120	1.160
LEU	C	9,551	914	1.174	1.107	1.160	1.181
LEU	N	13,765	1,391	2.398	2.175	2.408	2.446
LYS	H	13,706	1,659	0.452	0.391	0.416	0.446
LYS	HA	10,438	1,401	0.252	0.220	0.229	0.255
LYS	CA	10,890	1,208	0.911	0.882	0.916	0.954
LYS	CB	9,574	1,137	1.219	1.150	1.176	1.215
LYS	C	7,848	898	1.622	1.577	1.599	1.630
LYS	N	11,590	1,369	2.530	2.201	2.544	2.520
MET	H	3,644	1,340	0.481	0.440	0.470	0.489
MET	HA	2,838	1,120	0.295	0.243	0.255	0.274
MET	CA	3,176	1,054	1.042	0.994	1.065	1.125
MET	CB	2,775	981	1.544	1.472	1.544	1.542
MET	C	2,194	758	1.282	1.201	1.283	1.317
MET	N	3,187	1,152	3.003	2.637	2.958	3.001
PHE	H	7,030	1,568	0.515	0.467	0.493	0.509
PHE	HA	5,319	1,306	0.311	0.258	0.292	0.306
PHE	CA	5,726	1,180	1.267	1.123	1.219	1.337
PHE	CB	5,133	1,108	1.230	1.055	1.216	1.213
PHE	C	4,057	877	1.258	1.179	1.214	1.284
PHE	N	6,031	1,329	2.708	2.322	2.574	2.676
PRO	HA	5,998	1,311	0.291	0.242	0.249	0.272
PRO	CA	6,210	1,172	0.853	0.811	0.854	0.883
PRO	CB	5,493	1,100	1.070	1.026	1.080	1.089
PRO	C	4,289	856	1.205	1.088	1.161	1.259
PRO	N	125	39	16.207	16.635	16.790	15.906
SER	H	11,698	1,681	0.484	0.412	0.443	0.456
SER	HA	9,312	1,422	0.260	0.229	0.241	0.249
SER	CA	9,330	1,223	1.078	0.980	1.032	1.070
SER	CB	8,108	1,156	1.043	0.999	1.038	1.062
SER	C	6,734	914	1.209	1.143	1.218	1.245
SER	N	9,533	1,383	2.714	2.319	2.654	2.634
THR	H	10,575	1,642	0.492	0.431	0.462	0.477
THR	HA	8,152	1,388	0.278	0.241	0.254	0.264

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
THR	CA	8,284	1,202	1.200	1.131	1.168	1.196
THR	CB	7,261	1,137	2.102	2.079	2.071	2.141
THR	C	5,782	889	1.916	1.892	1.922	1.928
THR	N	8,868	1,356	3.154	2.926	3.382	3.323
TRP	H	2,495	1,137	0.571	0.514	0.536	0.551
TRP	HA	1,893	923	0.349	0.281	0.317	0.333
TRP	CA	1,914	851	1.407	1.286	1.439	1.570
TRP	CB	1,702	786	1.272	1.201	1.306	1.322
TRP	C	1,408	633	1.246	1.116	1.208	1.307
TRP	N	2,049	923	3.751	3.221	3.580	3.644
TYR	H	6,139	1,593	0.531	0.471	0.482	0.522
TYR	HA	4,665	1,339	0.325	0.271	0.298	0.319
TYR	CA	4,739	1,160	1.241	1.114	1.221	1.307
TYR	CB	4,148	1,083	1.299	1.164	1.270	1.296
TYR	C	3,360	864	1.213	1.140	1.177	1.220
TYR	N	5,052	1,313	2.747	2.430	2.767	2.810
VAL	H	13,232	1,655	0.482	0.430	0.469	0.496
VAL	HA	10,000	1,396	0.319	0.292	0.302	0.328
VAL	CA	10,719	1,222	1.131	1.127	1.139	1.149
VAL	CB	9,473	1,157	1.042	1.007	1.027	1.097
VAL	C	7,662	915	1.149	1.064	1.124	1.201
VAL	N	11,462	1,377	2.893	2.519	2.909	2.876

Table S19: Complete residue–backbone RMSE (ppm) for the four independently refitted static **ShiftTfmModel** source-contrast conditions on **RefDbTestLargeSet**. No first-retained-frame or trajectory model is included.

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
ALA	H	14,007	1,656	0.323	0.282	0.302	0.310
ALA	HA	10,471	1,400	0.174	0.142	0.153	0.169
ALA	CA	11,291	1,227	0.598	0.491	0.557	0.595
ALA	CB	10,013	1,159	0.693	0.633	0.675	0.713
ALA	C	8,064	910	0.869	0.815	0.868	0.915
ALA	N	12,049	1,386	1.788	1.553	1.846	1.788
ARG	H	8,937	1,620	0.349	0.305	0.339	0.348
ARG	HA	6,885	1,373	0.193	0.166	0.175	0.204
ARG	CA	7,051	1,198	0.775	0.684	0.719	0.734
ARG	CB	6,121	1,128	0.804	0.738	0.742	0.756
ARG	C	5,059	889	0.862	0.768	0.825	0.885
ARG	N	7,423	1,350	1.939	1.604	1.969	2.007
ASN	H	7,836	1,630	0.369	0.319	0.347	0.348
ASN	HA	6,063	1,376	0.180	0.152	0.168	0.183
ASN	CA	6,183	1,193	0.701	0.637	0.697	0.750
ASN	CB	5,522	1,122	0.744	0.684	0.765	0.807
ASN	C	4,356	881	0.848	0.777	0.867	0.872
ASN	N	6,490	1,345	2.042	1.762	2.075	2.113
ASP	H	11,068	1,647	0.342	0.306	0.332	0.346
ASP	HA	8,361	1,388	0.159	0.144	0.150	0.163
ASP	CA	8,892	1,210	0.662	0.596	0.631	0.688
ASP	CB	7,931	1,144	0.748	0.684	0.748	0.813
ASP	C	6,354	899	0.840	0.763	0.818	0.840
ASP	N	9,553	1,370	2.018	1.663	2.043	1.993
CYS	H	4,332	1,214	0.429	0.386	0.396	0.417
CYS	HA	3,774	1,033	0.274	0.225	0.260	0.267
CYS	CA	2,412	845	1.307	1.203	1.224	1.320
CYS	CB	2,115	782	2.229	1.950	2.254	2.299
CYS	C	1,757	632	1.042	0.969	1.046	1.068
CYS	N	2,832	957	2.501	2.017	2.395	2.555
GLN	H	7,491	1,564	0.321	0.289	0.302	0.310
GLN	HA	5,879	1,315	0.168	0.144	0.152	0.168
GLN	CA	6,045	1,181	0.649	0.580	0.633	0.673
GLN	CB	5,396	1,115	0.732	0.675	0.729	0.758
GLN	C	4,299	878	0.791	0.686	0.755	0.793
GLN	N	6,356	1,321	1.805	1.542	1.892	1.903
GLU	H	14,014	1,654	0.316	0.287	0.306	0.307
GLU	HA	10,649	1,398	0.155	0.137	0.144	0.158
GLU	CA	11,489	1,236	0.659	0.592	0.633	0.671
GLU	CB	10,150	1,163	0.687	0.622	0.677	0.647
GLU	C	8,429	920	0.816	0.766	0.768	0.810
GLU	N	12,179	1,397	1.804	1.519	1.824	1.902
GLY	H	13,701	1,679	0.343	0.321	0.350	0.348
GLY	HA	6,916	887	0.256	0.251	0.251	0.259
GLY	CA	10,685	1,229	0.566	0.484	0.566	0.600
GLY	C	7,570	913	0.969	0.898	0.948	0.972
GLY	N	11,299	1,389	2.150	1.798	2.097	2.098
HIS	H	4,061	1,411	0.405	0.362	0.398	0.407
HIS	HA	3,139	1,175	0.227	0.197	0.213	0.217

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
HIS	CA	3,301	1,077	1.006	0.899	0.964	1.051
HIS	CB	2,926	1,002	1.193	1.051	1.153	1.148
HIS	C	2,332	789	0.973	0.856	0.938	0.940
HIS	N	3,315	1,168	2.272	1.935	2.268	2.276
ILE	H	9,937	1,628	0.346	0.292	0.329	0.334
ILE	HA	7,514	1,374	0.215	0.170	0.199	0.213
ILE	CA	8,121	1,204	0.843	0.777	0.826	0.871
ILE	CB	7,194	1,142	0.797	0.723	0.784	0.854
ILE	C	5,817	896	0.833	0.743	0.791	0.871
ILE	N	8,575	1,350	1.996	1.813	2.057	2.167
LEU	H	16,096	1,681	0.340	0.294	0.319	0.327
LEU	HA	12,081	1,418	0.175	0.156	0.154	0.179
LEU	CA	13,137	1,228	0.589	0.551	0.592	0.601
LEU	CB	11,629	1,155	0.795	0.702	0.771	0.820
LEU	C	9,551	914	0.837	0.763	0.823	0.846
LEU	N	13,765	1,391	1.761	1.568	1.776	1.798
LYS	H	13,706	1,659	0.325	0.286	0.306	0.326
LYS	HA	10,438	1,401	0.174	0.148	0.157	0.182
LYS	CA	10,890	1,208	0.642	0.597	0.644	0.690
LYS	CB	9,574	1,137	0.713	0.639	0.669	0.709
LYS	C	7,848	898	0.831	0.752	0.791	0.844
LYS	N	11,590	1,369	1.869	1.586	1.888	1.866
MET	H	3,644	1,340	0.343	0.306	0.341	0.351
MET	HA	2,838	1,120	0.216	0.172	0.187	0.201
MET	CA	3,176	1,054	0.742	0.682	0.763	0.829
MET	CB	2,775	981	1.035	0.969	1.043	1.029
MET	C	2,194	758	0.909	0.822	0.904	0.954
MET	N	3,187	1,152	1.998	1.659	1.961	2.008
PHE	H	7,030	1,568	0.380	0.342	0.365	0.376
PHE	HA	5,319	1,306	0.229	0.187	0.218	0.230
PHE	CA	5,726	1,180	0.913	0.788	0.865	0.971
PHE	CB	5,133	1,108	0.887	0.733	0.881	0.879
PHE	C	4,057	877	0.918	0.828	0.868	0.953
PHE	N	6,031	1,329	2.048	1.708	1.934	2.018
PRO	HA	5,998	1,311	0.191	0.160	0.167	0.184
PRO	CA	6,210	1,172	0.572	0.531	0.581	0.601
PRO	CB	5,493	1,100	0.586	0.538	0.595	0.599
PRO	C	4,289	856	0.891	0.792	0.853	0.934
PRO	N	125	39	7.678	7.627	8.082	7.811
SER	H	11,698	1,681	0.337	0.290	0.313	0.319
SER	HA	9,312	1,422	0.178	0.156	0.164	0.173
SER	CA	9,330	1,223	0.764	0.674	0.724	0.761
SER	CB	8,108	1,156	0.634	0.589	0.632	0.645
SER	C	6,734	914	0.832	0.752	0.833	0.861
SER	N	9,533	1,383	1.997	1.712	1.959	1.954
THR	H	10,575	1,642	0.356	0.314	0.338	0.342
THR	HA	8,152	1,388	0.202	0.174	0.184	0.194
THR	CA	8,284	1,202	0.828	0.753	0.793	0.824
THR	CB	7,261	1,137	0.799	0.756	0.761	0.846
THR	C	5,782	889	0.858	0.785	0.843	0.883

continued

Residue	Atom	Assignments	Proteins	Plain	ORCA transfer	MOPAC	AIMNet2
THR	N	8,868	1,356	2.296	2.094	2.447	2.487
TRP	H	2,495	1,137	0.425	0.376	0.392	0.401
TRP	HA	1,893	923	0.266	0.206	0.241	0.250
TRP	CA	1,914	851	1.037	0.926	1.068	1.184
TRP	CB	1,702	786	0.916	0.838	0.935	0.962
TRP	C	1,408	633	0.915	0.801	0.885	0.987
TRP	N	2,049	923	2.344	1.835	2.204	2.260
TYR	H	6,139	1,593	0.388	0.345	0.356	0.389
TYR	HA	4,665	1,339	0.228	0.188	0.210	0.230
TYR	CA	4,739	1,160	0.901	0.782	0.890	0.944
TYR	CB	4,148	1,083	0.915	0.794	0.891	0.911
TYR	C	3,360	864	0.905	0.817	0.869	0.916
TYR	N	5,052	1,313	2.047	1.807	2.090	2.114
VAL	H	13,232	1,655	0.342	0.307	0.337	0.361
VAL	HA	10,000	1,396	0.234	0.216	0.225	0.245
VAL	CA	10,719	1,222	0.773	0.757	0.774	0.786
VAL	CB	9,473	1,157	0.681	0.644	0.666	0.725
VAL	C	7,662	915	0.839	0.745	0.804	0.897
VAL	N	11,462	1,377	2.135	1.839	2.148	2.164

Table S20: Complete residue–backbone MAE (ppm) for the four independently refitted static **ShiftTfnModel** source-contrast conditions on **RefDbTestLargeSet**. No first-retained-frame or trajectory model is included.